

Multiple Choice (10 marks)

1. Stars are often classified by their apparent brightness in the nighttime sky. Stars can also be classified in many other ways. Which of these is least useful in classifying stars?
 - (a) visible color
 - (b) composition
 - (c) surface texture
 - (d) temperature
2. What is the primary cause of rainstorms?
 - (a) Earth is heated by the Sun.
 - (b) Mountains direct moist air upward.
 - (c) Water forms droplets when cooled.
 - (d) Most of Earth is covered with water.
3. A river erodes its banks for thousands of years, and deposits the eroded material at its mouth. What two landforms did the river make?
 - (a) a hill and a delta
 - (b) a delta and a canyon
 - (c) a canyon and a dune
 - (d) a dune and a delta
4. A student sets up an experiment to learn which colors of light are refracted by a prism. Which method will the student most likely use to collect data?
 - (a) research
 - (b) interviews
 - (c) observation
 - (d) measurement
5. All elements found on the left side of the Periodic Table of the Elements have what properties in common?
 - (a) They are solids at room temperature.
 - (b) They don't conduct electricity.
 - (c) They are brittle and dull.
 - (d) They are radioactive.

True / False (10 marks)

Answer True or False

6. True or False: Most scientific discoveries are immediately accepted without any need for critical examination.
7. True or False: Using old materials can lead to increased spending on new resources.
8. True or False: Early photosynthetic life increased the level of oxygen in Earth's atmosphere.
9. True or False: Water's ability to freeze is the most important property for organisms to thrive on Earth.
10. True or False: The kidneys remove dissolved waste molecules from the blood after cell metabolism.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss how the unique environmental challenges of living above the Arctic Circle would influence the development of specific inventions. Provide examples and analyze their significance.
12. During a nature walk, what specific observations about grasshoppers can your science class make? Discuss the significance of these observations in understanding their behavior and environment.

End of Paper